ESP32 Radar Scanner

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Overview

A radar scanner built on an ESP32 microcontroller that uses an ultrasonic sensor mounted on a servo motor to sweep 0–180° and detect objects. Distances are displayed in real time on a 1.8" TFT screen as a radar-style visualization. A buzzer sounds when an object is detected closer than 25cm.



Components

- ESP32 development board
- HC-SR04 ultrasonic distance sensor
- SG90 servo motor
- 1.8" ST7735 TFT display (128x160)
- Active buzzer
- 220Ω resistor (buzzer current limiting)
- 1kΩ and 2.2kΩ resistors (voltage divider for ECHO pin)
- Breadboard and jumper wires
- Power bank (5V USB)

Pin Connections

- Servo signal — GPIO 13
- HC-SR04 TRIG — GPIO 33
- HC-SR04 ECHO — GPIO 32 voltage divider)
- Buzzer positive — GPIO 12 (220Ω resistor)
- TFT SCL — GPIO 19
- TFT SDA — GPIO 23
- TFT RES — GPIO 4
- TFT DC — GPIO 2
- TFT CS — GPIO 15
- TFT VCC — 3.3V
- HC-SR04 VCC — VIN (5V)
- Servo VCC — VIN (5V)
- All GND — GND

Voltage Divider (ECHO Pin Protection)

The HC-SR04 outputs 5V on the ECHO pin which would damage the ESP32 3.3V GPIO. A voltage divider brings it down to about 3.4V:

- 1kΩ resistor between ECHO and GPIO 32
- 2.2kΩ resistor between GPIO 32 and GND

Display Layout

- Top bar: “RADAR SCANNER” title with green underline
- Middle: radar with 4 rings representing 1m, 2m, 3m, 4m range. Grid lines every 30°
- Bottom: current angle in degrees and distance in cm. The detection dot is red.

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How It Works

The servo sweeps from 0° to 180° and back continuously. At each degree the HC-SR04 fires an ultrasonic pulse and measures the time for the echo to return, converting it to a distance in centimeters.

The TFT displays a half-circle radar with four range rings, a green sweep line following the servo angle, and a blue dot at the detected object's position. Angle and distance are shown as text at the bottom of the screen.

When a detected object is closer than 25cm the active buzzer turns on continuously and turns off when the object moves beyond that threshold.

Code Structure

measureDistance()

Triggers the HC-SR04 by sending a 10 microsecond pulse on the TRIG pin, then listens on the ECHO pin for the reflection. The time the echo takes to return is converted to centimeters using the speed of sound. If no echo returns within 30ms the function returns 400cm as a maximum range value. The result is also clamped so nothing above 400 can slip through.

drawTitle()

Runs once at startup and draws the “RADAR SCANNER” text and green underline at the top of the screen. It is called only in setup and never again, which means the radar redrawing below it never touches or flickers the title.

drawRadar()

Called every loop cycle. It clears the radar area, redraws the four range rings and the grid lines every 30°, then draws the green sweep line at the current servo angle using trigonometry to convert degrees to pixel coordinates. If a valid distance is detected a blue filled circle is drawn at the mapped position between the center and the edge of the radar. The bottom text showing angle and distance is also updated here.

handleBuzzer()

Checks the current distance and simply turns the buzzer on with digitalWrite HIGH if the object is closer than 25cm, or off with LOW if it is beyond that. Because it uses an active buzzer the tone is fixed internally — the code only controls whether it is on or off.

loop()

It ties everything toghether.